



Vijay Academy Presents



Savitribai Phule Pune University

Artificial Intelligence (AI)

TE Computer Engineering

Endsem Exam Strategy (Unit 3-6)

Most IMP Topics + PYQs

What You Get : Strategy For Exam :

- Simple Explanation
- PPT + Notes
- PYQs + Most IMP Questions



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UNIT 3 — Adversarial Search and Games

MOST IMPORTANT TOPICS (Asked 3+ times)

1. Alpha-Beta Tree Search & Cutoff (Asked 5 times)

Explain Alpha-Beta tree search and cutoff procedure with example (3)

How alpha-beta cut-offs improve minimax performance (2)

2. Minimax Algorithm (Asked 4 times)

Write Minimax Search Algorithm for two players (2)

Explain Min Max and Alpha Beta pruning algorithm with example (1)

Explain how Minimax and alpha-beta change for non-zero-sum games (1)

3. Constraint Satisfaction Problem (CSP) (Asked 6 times)

Define CSP, state types of consistencies, solve SEND+MORE=MONEY (2)

Define CSP, state types of consistencies, solve BASE+BALL=GAMES (2)

What is CSP, explain with example (1)

Define CSP, explain Map Coloring Problem, formulate as CSP (1)

4. Backtracking & N-Queens Problem

(Asked 5 times)

Explain backtracking and constraint propagation, solve N-queens (3)

What is backtracking, explain with N-queens (2)

5. Monte Carlo Tree Search (MCTS)

(Asked 2 times)

Short note on MCTS and its limitations (1)

Explain MCTS with all steps and demonstrate with example (1)



MEDIUM IMPORTANT TOPICS:

6. Stochastic Games vs Partially Observable Games (Asked 3 times)

Differentiate stochastic and partial observable games with examples (2)

Compare stochastic and partially observable games (1)

7. Constraint Propagation (Asked 2 times)

Explain constraint propagation, node consistency and arc consistency (1)

Issues in solving CSP efficiently, explain solutions (1)

8. Game Theory (Asked 2 times)

Define Game Theory, components, draw game tree for tic-tac-toe (1)

Define Game Theory, differentiate stochastic and partial games (1)

9. Tic-Tac-Toe using AI (Asked 2 times)

How AI technique is used to solve tic-tac-toe problem (2)

10. Graph/Map Coloring Problem (Asked 2 times)

Explain graph coloring problem with example (1)

Define CSP, explain Map Coloring, formulate as CSP (1)



LEAST IMPORTANT TOPICS

- 11. Limitations of Game Search Algorithms — asked once**
- 12. Optimal Decisions in Games (theory only) — Never asked alone**
- 13. Partially Observable Games (alone) — Never asked alone**



UNIT 4 — Knowledge

MOST IMPORTANT TOPICS (Asked 3+ times)

1. Wumpus World

(Asked 5 times)

Explain Wumpus World environment with PEAS description (3)

Explain Knowledge Based Agent with Wumpus World (2)

2. Knowledge Based Agent

(Asked 5 times)

Explain Knowledge Based Agent with Wumpus World (3)

What is an Agent, name 5 agents, explain Knowledge Based Agent (2)

3. First Order Logic — FOL (Sentences & Quantifiers)

(Asked 5 times)

Write sentences in FOL using quantifiers — All birds fly, Some boys play cricket etc. (3)

Represent facts in predicate/FOL logic — triangle axioms (2)



4. Propositional Logic vs First Order Logic

(Asked 4 times)

Compare propositional and predicate logic (2)

Knowledge representation using propositional logic, compare PL and FOL (2)

5. Inference Rules in Propositional Logic

(Asked 3 times)

List and explain inference rules in propositional logic with examples (2)

Explain different inference rules in FOL with suitable example (1)

6. Syntax and Semantics of FOL

(Asked 3 times)

Explain syntax and semantics of First Order Logic in detail (2)

Short notes on Syntax and Semantics (1)



MEDIUM IMPORTANCE TOPICS

7. Knowledge Engineering Process (Asked 3 times)

List and explain steps of knowledge engineering process (2)

Knowledge Engineering Process in First Order Logic (1)

8. Knowledge Representation Structures (Asked 2 times)

Explain knowledge representation structures and compare them (2)

9. Propositional Theorem Proving / Entailment (Asked 2 times)

Detail the algorithm for deciding entailment in propositional logic (1)

Show sentences are valid or not — propositional logic (1)

10. Agents Based on Propositional Logic (Asked 1 time)

Explain knowledge based agent with architecture diagram, state significance of inference engine (1)

LEAST IMPORTANT TOPICS

11. Logical Agents (theory only) — Never asked alone

12. Effective Propositional Model Checking — Never asked alone

UNIT 5 — Reasoning

MOST IMPORTANT TOPICS (Asked 3+ times)

1. Forward Chaining & Backward Chaining (Asked 6 times)

Explain Forward and Backward Chaining with example (3)

Explain Forward Chaining algorithm with example (1)

Explain Backward Chaining algorithm with example (1)

Forward and Backward Chaining with properties, advantages, disadvantages (1)

2. Unification Algorithm (Asked 4 times)

Explain Unification algorithm with example (2)

Unification in FOL — solve stepwise $p(x, g(x)) = f(\text{prime}, f(\text{prime}))$ (1)

Illustrate unification algorithm to prove resolution (1)

3. Resolution (Asked 4 times)

Explain Resolution, solve using resolution algorithm with graph (1)

Short note on Resolution and Unification (1)

Explain Resolution, procedure to convert to CNF (1)

Illustrate unification to prove resolution (1)



4. FOL Inference — Quantifiers

(Asked 4 times)

Explain FOL inference for Universal Generalization, Universal Instantiation, Existential Instantiation, Existential Introduction (4)

5. Ontological Engineering

(Asked 3 times)

Explain Ontological Engineering with categories, objects and model (3)



MEDIUM IMPORTANCE TOPICS

6. Knowledge Representation & Reasoning Systems (Asked 2 times)

Explain knowledge representation structures and compare them (1)

Reasoning systems for categories — Semantic Network and Description Logic (1)

7. Reasoning with Default Information (Asked 2 times)

Explain reasoning with default information (2)

8. Categories and Objects (Asked 2 times)

Describe Ontological Engineering w.r.t Categories, Objects and Model (1)

Short notes on Categories and Objects (1)

9. Propositional vs FOL Inference (Asked 1 time)

Propositional vs First Order Inference (1)

10. Reasoning Patterns in Propositional Logic (Asked 1 time)

What are reasoning patterns in propositional logic, explain in detail (1)

LEAST IMPORTANT TOPICS

11. Mental Objects and Modal Logic — Never asked alone

12. Inference in First Order Logic (theory only) — Never asked alone



UNIT 6 — Planning

MOST IMPORTANT TOPICS (Asked 3+ times)

1. Classical Planning & Algorithms (STRIPS) (Asked 5 times)

Explain Goal Stack Planning / STRIPS algorithm with example (3)

Explain algorithm for classical planning with example (2)

2. AI Components & AI Architecture (Asked 5 times)

Explain AI components and AI architecture (5)

3. Planning vs Problem Solving (Asked 4 times)

Explain with example how planning is different from problem solving (4)

4. Planning in Nondeterministic Domains (Asked 3 times)

Explain planning in nondeterministic domain (3)

5. Limits of AI & Future of AI (Asked 4 times)

Explain limits of AI and future opportunities with AI (2)

What is AI, explain scope of AI in all walks of life and future opportunities (2)



MEDIUM IMPORTANCE TOPICS

6. Analysis / Types of Planning Approaches (Asked 3 times)

Analyse various planning approaches in detail (2)

What are types of planning, explain in detail (1)

7. Hierarchical Planning (Asked 2 times)

Explain Hierarchical Planning with relevant example (1)

Write note on hierarchical task network planning (1)

8. Ethics of AI (Asked 2 times)

Discuss AI and its ethical concerns (1)

Explain limitations of AI and ethical concerns (1)

9. Heuristics for Planning (Asked 1 time)

Explain importance of planning algorithm for classical planning (1)

10. Time, Schedules and Resources (Asked 1 time)

Explain terms for time and schedule from perspective of temporal planning (1)

11. Blocks World (Asked 1 time)

Explain the Blocks World in detail (1)

LEAST IMPORTANT TOPICS

12. Automated Planning (theory only) — Never asked alone

13. State Space Planning — Never asked alone



Thank You!



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